

The Human Advantage

Making Better QA Decisions with AI in the Loop

If AI can generate test cases, write automation, analyze logs, summarize defects, and even suggest root causes — what is left for QA professionals?

The most valuable part: Judgment.

Introduction

- ❖ *Human first*
- ❖ *Love testing & breaking things, solving problems*
- ❖ *Gardening & DIYs when free*
- ❖ *Love to volunteer!*
- ❖ *Married, Dad of 3 kids*
- ❖ *Team Lead QA @Encova Insurance, COSQAM board member*
- ❖ *Reading, Listening & Biking*



Agenda

1

Why this talk, why now? / The Reality Check

2

The Evolution of QA

3

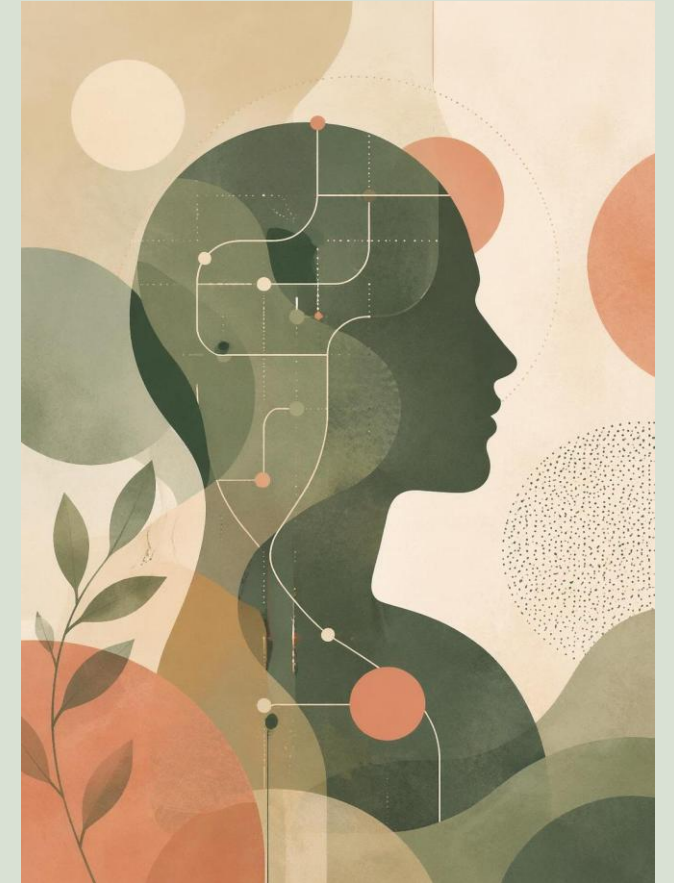
The Cognitive QA Toolkit – 5 mental models

4

Personal Growth Strategies

5

Summary & Challenge



Why This Talk? Why Now?

79% of Ohio organizations now call AI a high priority (up from 48%)

57% of Ohio organizations have generative AI in production or fully integrated (up from 32%)

No measurable ROI dropped from **33% to 16%**

The #1 barrier flipped from "lack of clear business cases" to "limited technical expertise" — talent is the new bottleneck

The Big Question:

What makes QA professionals IRREPLACEABLE in the age of AI?

Are you someone who runs the tests? ----- Or someone who OWNS Quality?

The answer defines your career in the age of AI.

The Reality Check

AI is changing QA — but not replacing it.

TRADITIONAL QA



AI-ASSISTED QA



AI does

- Writes tests
- Generates data
- Summarizes logs
- Explains code
- Helps debug

AI does NOT

- Understand organizational risk
- Understand customer emotions
- Own production incidents
- Make accountability decisions

"AI increases capability. Humans provide responsibility & judgement."

The Evolution of QA – Use AI as a Thinking Partner

How QA roles are changing.

BEFORE	With AI
Execute tests	Evaluate risk
Verify requirements	Challenge assumptions
Find bugs	Prevent failures
Test features	Protect customer outcomes
Report defects	Influence decisions

 AUTOPILOT (Risky)

- Blindly trust AI output
- Skip validation
- Lose domain context

 CO-PILOT (Effective)

- AI generates, you validate
- Add domain expertise
- Own the quality decisions

*"The future QA engineer is not the best test executor — it's the best quality thinker."
AI accelerates your thinking – it doesn't replace it*

The Cognitive QA Toolkit

5 mental models every QA professional should master

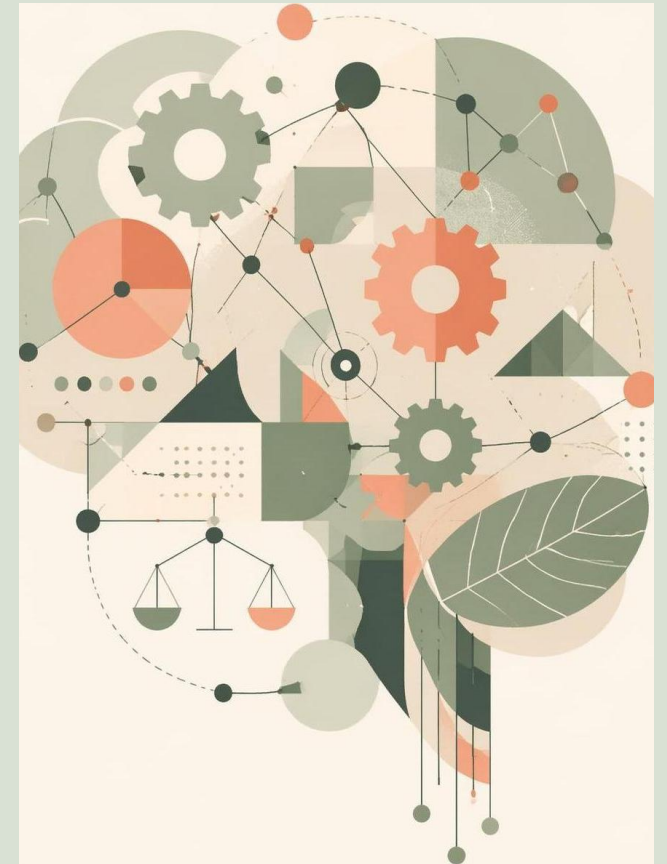
1 ***Risk Based Thinking***

2 ***Systems Thinking***

3 ***Healthy Skepticism***

4 ***Inversion Thinking***

5 ***Prompt Patterns***



Risk-Based Thinking

Great testers think in risk, not test cases.

ASK THE QUESTION

"If you only had one hour to test before production, what would you test?"

Login page failure → high impact

THE RISK FORMULA

$$\text{Risk} = \text{Probability} \times \text{Impact}$$

Typo in FAQ page → low impact

HIGH
Criticality

STANDARD
Test thoroughly

CRITICAL
Test first, automate

LOW
Criticality

AS NEEDED
Spot check

MONITOR
Automate in CI/CD

Low Frequency ↔ High Frequency

Before testing, ask:

1 What can go wrong?

2 How likely is it?

3 Who gets hurt?

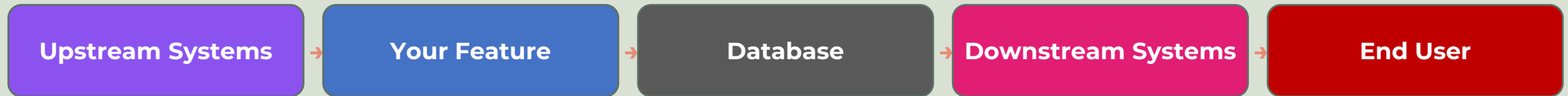
4 How expensive is it?

AI PROMPT

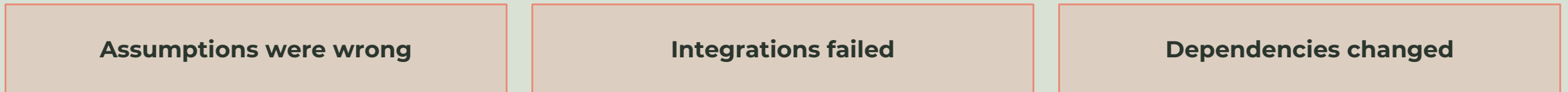
"Act as a risk analyst and identify the top 10 failure scenarios for this feature." → Human decides which scenarios matter.

Systems Thinking

Bugs rarely live alone.



Most production incidents happen because:



QA professionals must understand:



"What happens after this screen? What is the big picture?" — Great QA asks that repeatedly.

Product Skepticism

Skeptical, not cynical.

CYNICAL QA

"That won't work."

CURIOUS QA

"What happens if...?"

WHAT IF...

The API is slow?

The member has multiple policies?

The user enters emoji?

Payment succeeds but confirmation fails?

THE CURIOUS SKEPTIC FRAMEWORK

What assumptions exist? · What dependencies exist? · What could break? · What haven't we considered?

✗ ADVERSARIAL

- "I'm here to break everything"
- Bug reports feel like accusations
- Team avoids sharing with QA

✓ HEALTHY

- "I trust, but I verify"
- Question assumptions, not people
- Team seeks QA input early

Inversion Thinking for QA

"All I want to know is where I'm going to die, so I'll never go there." — Charlie Munger

How does this work?



How would this FAIL?

What should happen?



What SHOULDN'T happen?

Is this correct?



Under what conditions is this WRONG?

Did the test pass?



What did the test NOT cover?

Is this requirement clear?



How could this be MISinterpreted?

Prompt Patterns Every QA Should Know

Good prompts create better thinking & better AI responses.

01 GENERATE (Persona)

Act as a QA engineer. Generate happy path, edge case, negative, security, and usability test scenarios for this feature.

02 CHALLENGE (Chain-of-Thought)

Challenge this requirement. First analyze, then Identify ambiguity, assumptions, and missing acceptance criteria.

03 INVESTIGATE

Act as a production incident investigator. List possible root causes.

04 EMPATHIZE (Perspective)

Generate test scenarios from a customer perspective.

Personal Growth Strategies

1

Communication

2

Build Trust

3

The T-Shape Growth

4

Learn in Public

5

Build Resilience



Communication is a QA Superpower

Finding bugs doesn't create value — communicating RISK does.

✗ Gets Ignored

Title: Login doesn't work

Description: It just fails.

Steps: Go to login page.

✓ Gets Fixed Fast

Title: Mobile login returns 500 for special char passwords

Impact: ~12% of users affected

Recommendation: Add input sanitization

R

Risk

What's at stake?

I

Impact

Who is affected?

S

Scope

How widespread?

K

Know

What do/don't we know?

THE P-I-L-R FRAMEWORK

P

Problem

What is broken

I

Impact

Who is affected

L

Likelihood

How often it happens

R

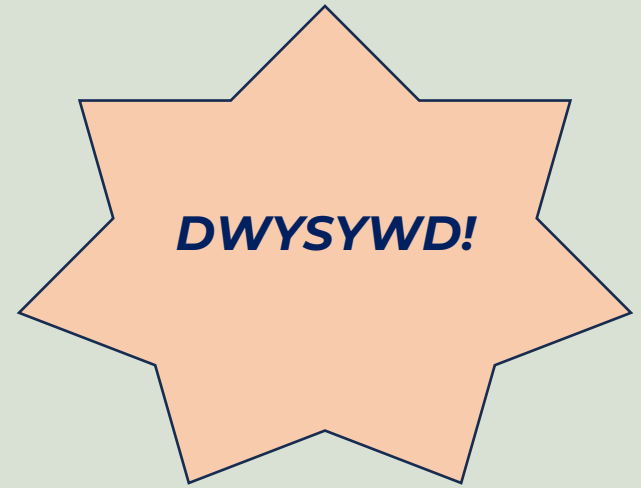
Recommendation

What to do next

This helps leaders make decisions quickly.

Building Trust

Trust is the currency of QA. Become the trusted advisor



TRUST COMES FROM

Consistency

Accuracy

Transparency

Follow-through

People trust QA professionals who:

- ❖ Admit uncertainty
- ❖ Escalate appropriately
- ❖ Explain reasoning
- ❖ Focus on outcomes
- ❖ Quantifies the Risk
- ❖ Provides Risks + Recommendations
- ❖ DWYSYWD

"The goal is not to be right. The goal is to help the team make better decisions."

The T-Shape

Deep QA expertise + broad cross-functional knowledge

- ❖ Domain (Insurance)
- ❖ Development Practices
- ❖ DevOps & CI/CD
- ❖ Business Acumen
- ❖ AI & ML Fundamentals

DEEP
QA

Action Items:

- ☐ Shadow a developer
- ☐ Learn SQL for test data
- ☐ Understand CI/CD pipeline
- ☐ Read business requirements
- ☐ Take an AI fundamentals course

Learn in Public

Share what you learn to grow faster

Blog

Write about testing challenges

Speak

Present at meetups & conferences

Mentor

Guide junior testers

Share

Post insights on LinkedIn & Slack

Document

Create team knowledge bases

My Journey

2023

Opensource Automation Tool Comparison

2024

Prompt Engineering Deep Dive

2025

Prompt Engineering Deep Dive X 2

2026

The Human Advantage (Today!)

"You don't have to be an expert — just one step ahead."

Build Resilience

Adapting and thriving through change

Embrace Change

See AI as tool.
Adapt proactively.

Control Your Circle

Focus on skills,
not headlines.

Seek Feedback

Ask for input.
Growth needs it.

Reflect Regularly

Journal daily.
Build awareness.

Focus is Power!

Micro-Habit: The 2-Minute Reset

Close eyes, 5 deep breaths, ask: What's the most important thing I should be doing next?

Personal Growth Habits

Habits that build better judgment.

DAILY (EOD)

- ❖ Read one bug report
- ❖ Read one production incident
- ❖ Ask one "What am I missing?" question
- ❖ Learn one thing
- ❖ One win to celebrate
- ❖ Tomorrow's top priority
- ❖ Thought experimentation

WEEKLY (EOW)

- ❖ Review escaped defects
- ❖ Review customer complaints
- ❖ Reflect on decisions
- ❖ What patterns emerged?
- ❖ Feedback given/received
- ❖ Learning goal for next week
- ❖ Process / habitual adjustments

MONTHLY (EOM)

- ❖ Learn one new AI prompt pattern
- ❖ Learn one business process
- ❖ Teach someone something
- ❖ Work on a new feature
- ❖ Challenge your thoughts
- ❖ Review the decisions made

Pre-Release Checklist (Human Edition)

7 checkpoints only humans can assess

- ☐ Have I tested the happy path AND the unhappy path?
- ☐ Have I considered the user who doesn't read instructions?
- ☐ Have I checked what happens at system boundaries?
- ☐ Have I tested with realistic data, not just test data?
- ☐ Have I communicated risks clearly to stakeholders?
- ☐ Have I verified the fix didn't break something else?
- ☐ Would I be comfortable if MY NAME was on this release?

Building Intuition Through Consistency

Small daily habits = massive professional growth

"Intuition isn't magic — it's pattern recognition from repeated practice."

Week 1

Feels
awkward

Month 1

Becoming
automatic

Month 3

Seeing
patterns

Month 6

Predicting
defects

Year 1

Trusted
intuition

What separates great testers from good ones isn't talent — it's habits.

The Human Advantage — Summary

Five themes for the modern QA professional

AI Amplifies

Use as co-pilot.
Validate. Own.

Mental Models

Think in risk,
systems, inversion.

Communication

Speak in R.I.S.K.
Be the advisor.

Growth Mindset

T-shape. Learn
in public.

Daily Habits

Small habits
compound.

The Human Advantage is the COMBINATION:

Think deeply • Use AI wisely • Communicate clearly • Grow continuously • Practice consistently

Your Challenge: Starting Monday

Five actions for the next 7 days - Consistency beats intensity.

1 Use ONE prompt pattern to generate test cases — then validate every one

2 Apply the R.I.S.K. framework to your next status report

3 Rewrite ONE bug report: Context + Impact + Steps + Evidence + Recommend

4 Start your daily habit stack — just 2 items, morning and evening

5 Share one thing you learned this week with a colleague

Thank you!

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Quality Assurance Team Lead at Encova Insurance

❖ *Please scan the QR to provide feedback*



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